

FILTER DESIGN TEST CASE

BAW (Bulk Acoustic Wave) - Band 2 (1900 MHz) for WiFi 6E Coexistence Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0056
Revision:	Rev B.6
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Created Date:	2025-04-06
Last Modified:	2025-05-16
Status:	CONDITIONAL PASS
Priority:	P2 - High

1. OBJECTIVE

This document describes the design, simulation, and characterization of a BAW (Bulk Acoustic Wave) filter targeting Band 2 (1900 MHz) for WiFi 6E Coexistence Filter. The filter is designed on Langasite substrate with a target center frequency of 2821.5 MHz and bandwidth of 132.8 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the WiFi 6E Coexistence Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: BAW (Bulk Acoustic Wave)
- Frequency Band: Band 2 (1900 MHz)
- Application: WiFi 6E Coexistence Filter
- Substrate: Langasite
- Package: WLP (Wafer Level Package)
- Design Tool: Keysight ADS 2024
- Design Center: Osaka Design Center

This specification applies to all variants of the BAW (Bulk Acoustic Wave) design targeting Band 2 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	2821.5 MHz
Bandwidth (3 dB):	132.8 MHz
Insertion Loss Target:	≤ 2.9 dB
Return Loss Target:	≥ 13.0 dB
Out-of-Band Rejection:	≥ 23.0 dB
Isolation (Duplexer):	≥ 23.0 dB
Q Factor:	1016
Temperature Coefficient:	-24.3 ppm/°C
Die Size:	1.7 x 0.7 mm
Wafer Size:	6-inch
Number of Resonators:	7
Electrode Material:	W

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Keysight ADS 2024 and load the BAW (Bulk Acoustic Wave) template project.
2. Define the substrate stack: Langasite with specified layer thicknesses.
3. Set design targets: center freq 2821.5 MHz, BW 132.8 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.9 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (WLP (Wafer Level Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, Langasite).
10. Perform on-wafer probing using Keysight N9041B UXA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 8464 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.9 dB
- Return loss in passband shall be >= 13.0 dB
- Out-of-band rejection at +/- 199 MHz offset >= 23.0 dB
- Group delay variation across passband shall be <= 9 ns
- Temperature drift over -40°C to +85°C shall be <= 8.6 MHz
- Power handling shall be >= +32 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 97%

6. TEST RESULTS

Measurement Date: 2025-05-16
Devices Tested: 18
Insertion Loss (meas): 2.61 dB
Return Loss (meas): 15.3 dB
Rejection (meas): 25.6 dB
Isolation (meas): 28.1 dB
Q Factor (meas): 1016
Group Delay Variation: 5.5 ns
Power Handling: +32 dBm
Die Yield: 95.0%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The BAW (Bulk Acoustic Wave) design demonstrated acceptable performance for Band 2 (1900 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Temperature stability was within specification across full range.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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